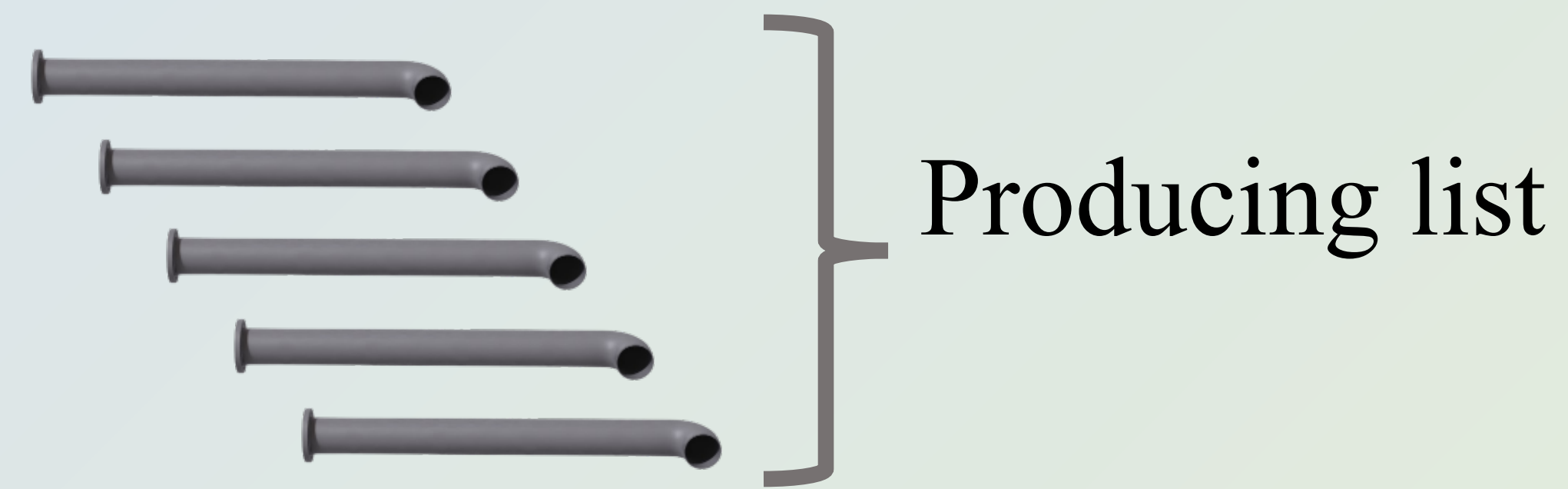
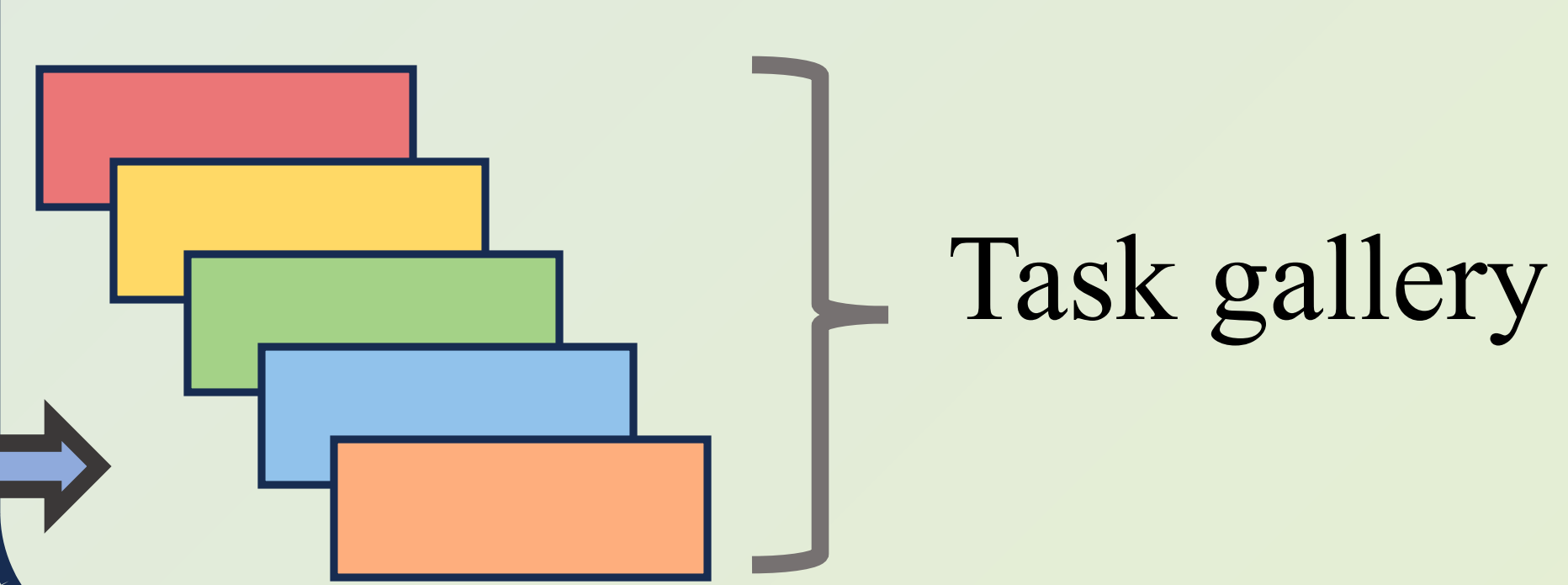


Real time task planning and allocation (TPA)

Step 1: Product selection



Step 2: Task planning

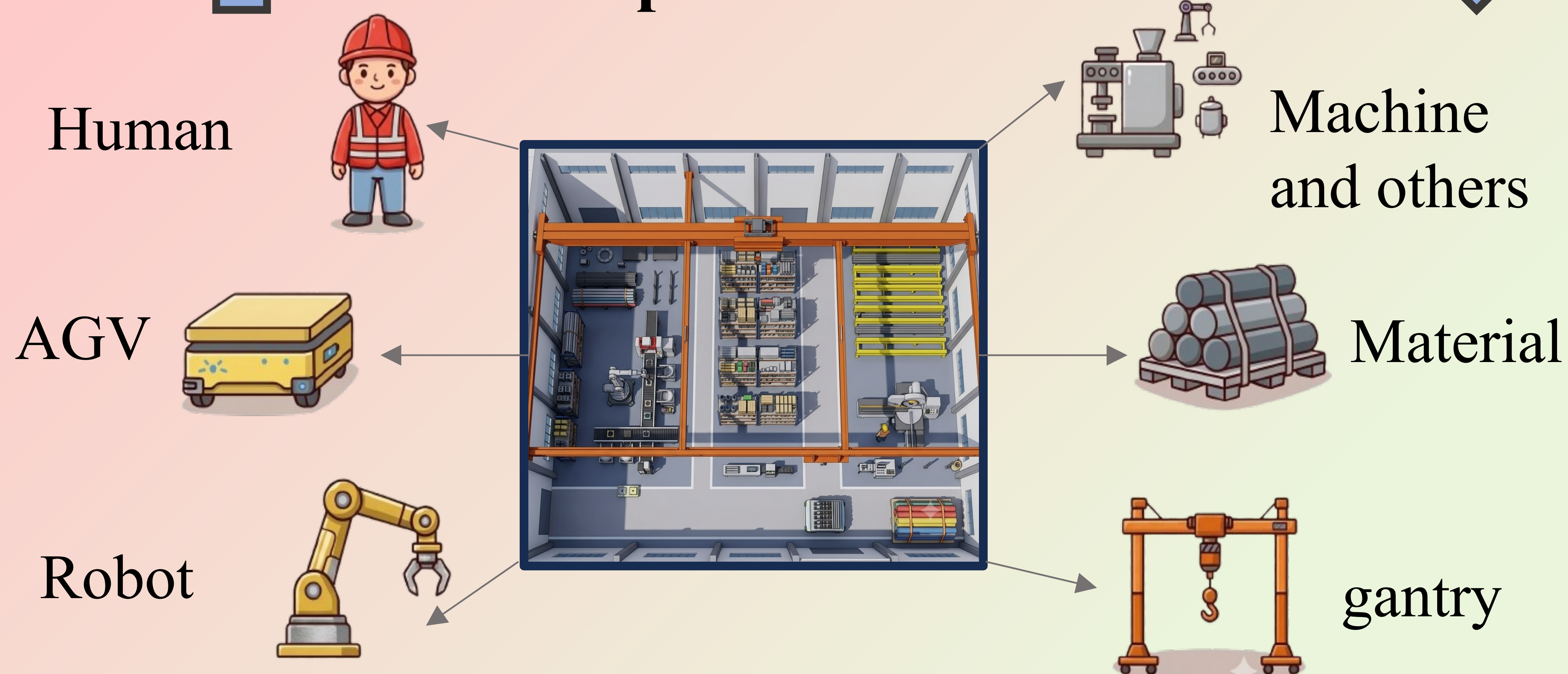


Step 3: Task allocation

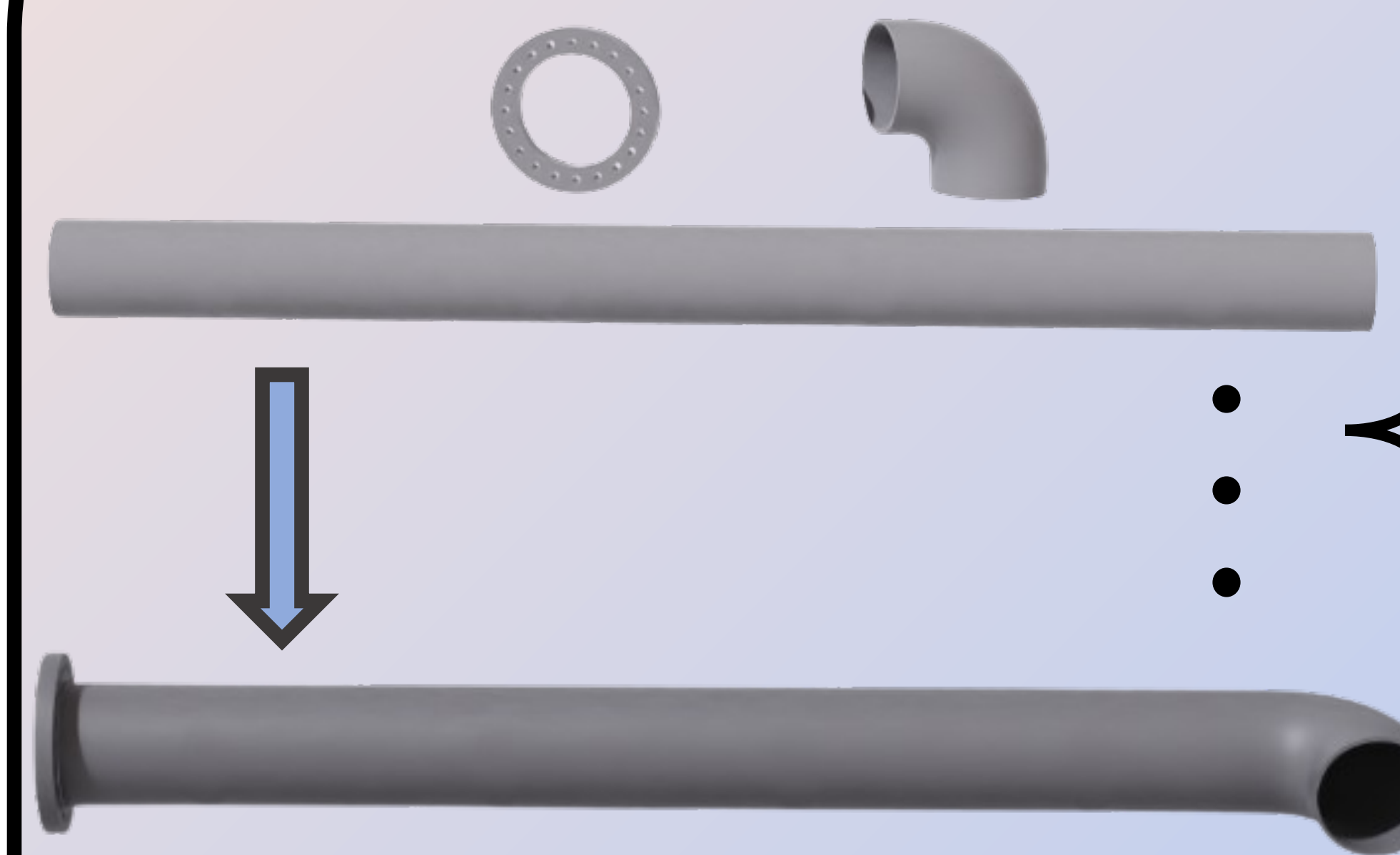


At time t , for product i , we plan task j and allocate task j to human m and robot n .

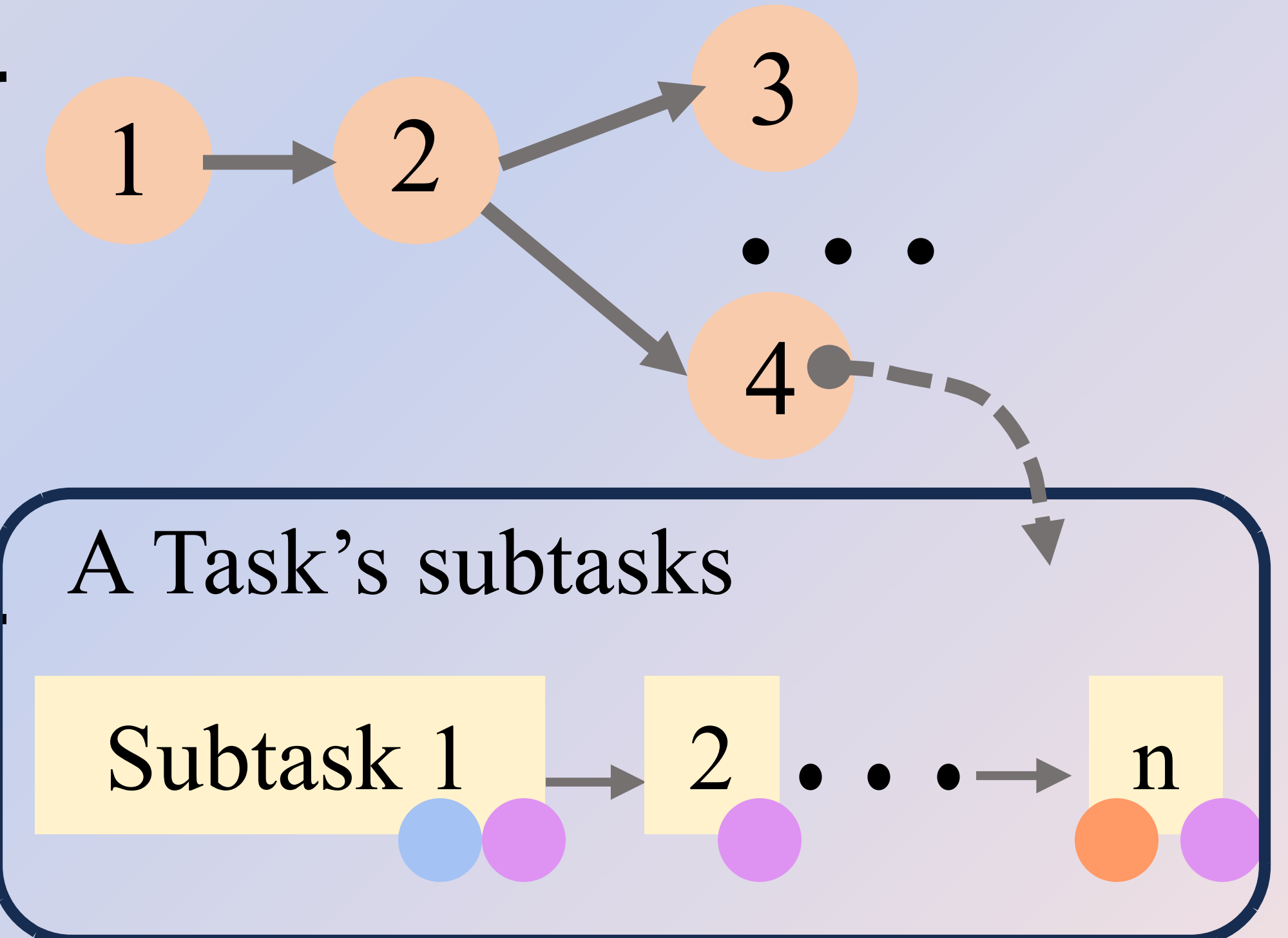
MC production env



Product process

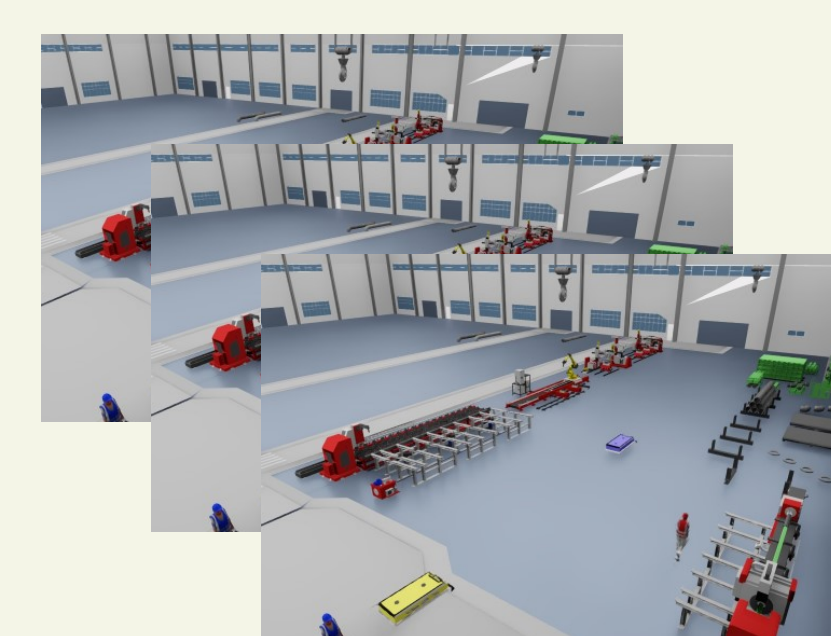


Task dependency graph



Participant: ● Human ● Robot ● Machine

Data generation and human state estimation training



Images



Text data



Dataset

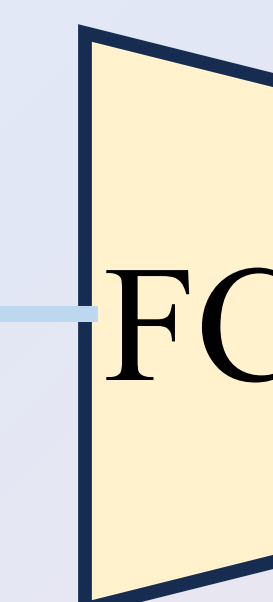
Automatic labeling data

Multi-camera view

Human ID model

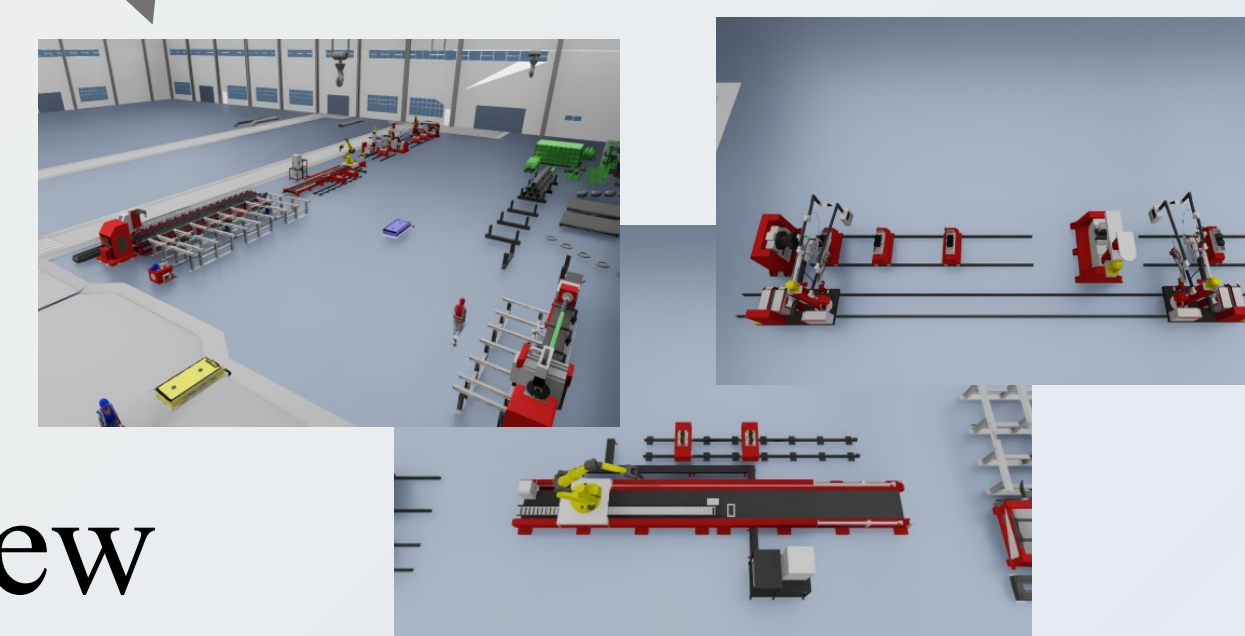


Resnet18



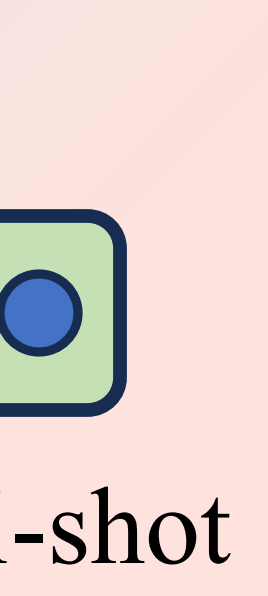
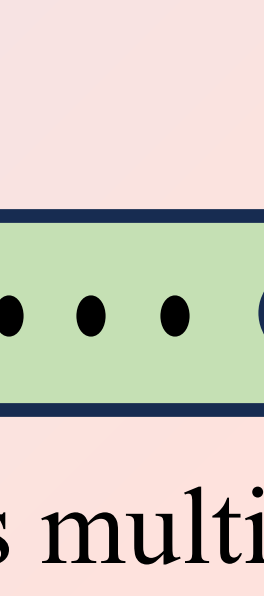
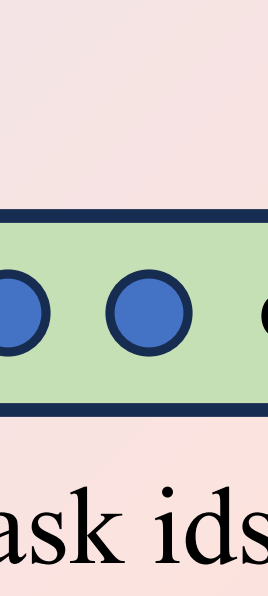
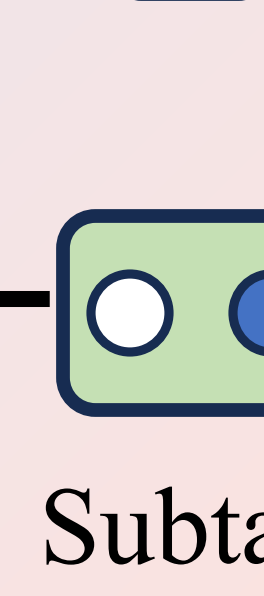
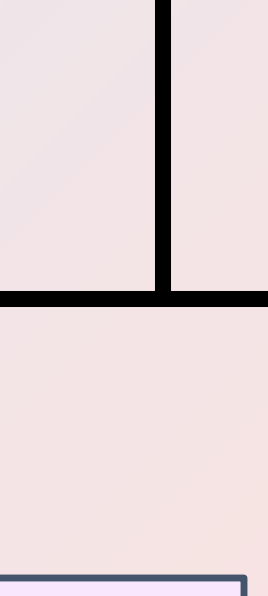
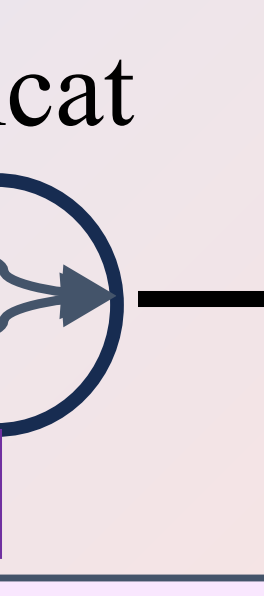
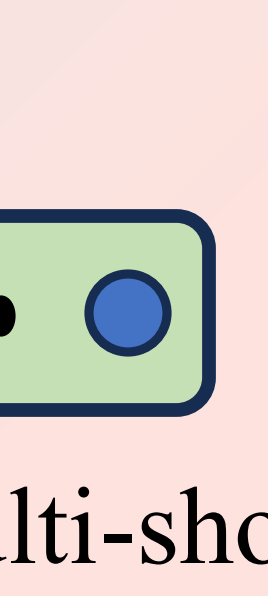
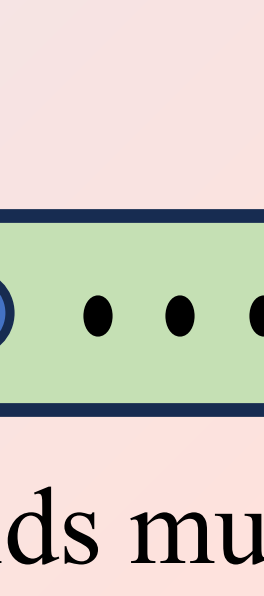
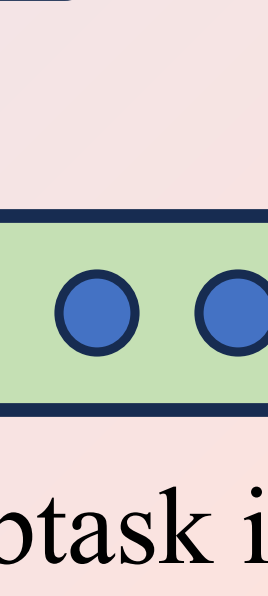
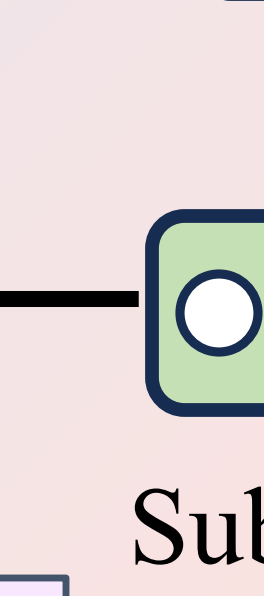
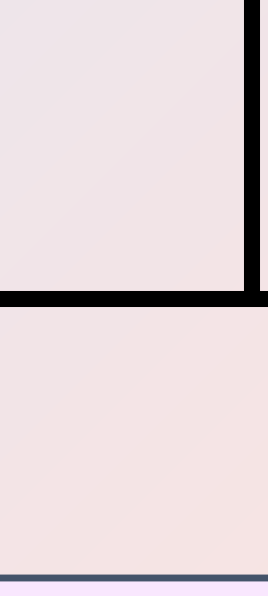
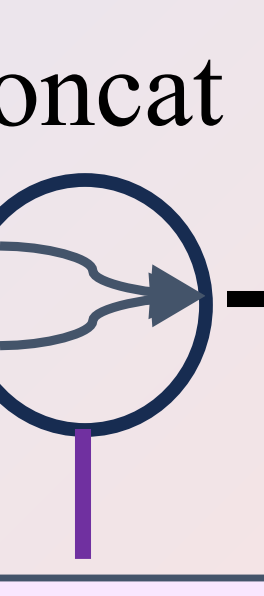
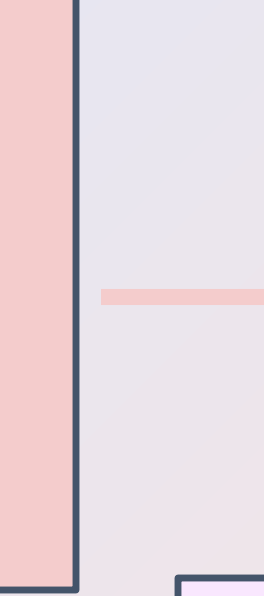
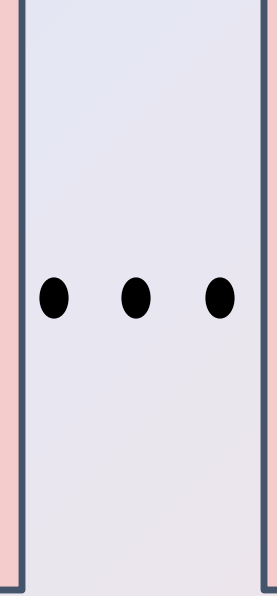
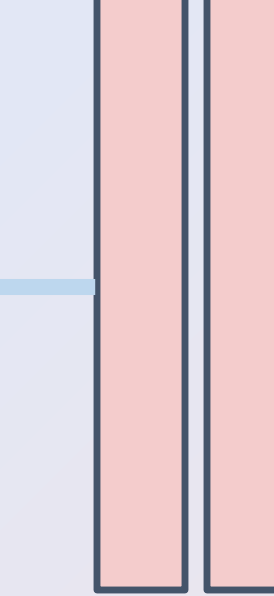
Human ids multi-shot

Human subtask model



Resnet18

Cameras encoding



Task embedding

Subtask ids multi-shot

Subtask done